PHENIKAA UNIVERSITY - SCHOOL OF COMPUTING

Group Submission Software Architecture

Project Title: Kanban Board Application

Group Submission: N02 – Group 2

Name	Student ID	Position Title
Trần Thành Long	23010070	Code Backend – Frontend – Review Final

Class: CSE703110-1-2-25(N02)

Instructor in charge : Vũ Quang Dũng

Hà Nội - Feb 7, 2026

Full-Stack Implementation & Engineering

Project: Kyte - Kanban Board Application
Course: Software Architecture
Student: Trần Thành Long (23010070)
Role: Full-stack Developer
Class: N02 - Group 2
Date: Feb 07, 2026

1. Executive Summary

As the Full-stack Developer for the Kyte project, I was responsible for the end-to-end implementation of the entire system. My primary focus was on transforming the architectural designs into a high-performance, real-time collaborative platform. I developed the complete backend infrastructure using Bun and Elysia, engineered the database schema with Drizzle ORM, and built the interactive React 19 frontend. A major part of my contribution was the implementation of the core domain logic including real-time event orchestration and fractional indexing algorithms and the authoring of a robust suite of over 130 automated tests to ensure system reliability and data integrity.

2. Technical Contribution Scope

My implementation work covered the full depth of the Kyte stack:

- **Backend Infrastructure:** Developed a modular monolith API server using ElysiaJS and Bun runtime.
- **Frontend Engineering:** Built a reactive, type-safe SPA using React 19, TanStack Router, and TanStack Query.
- **Real-Time Orchestration:** Implemented the system-wide Event Bus and WebSocket Bridge for instant synchronization.
- **Algorithmic Implementation:** Engineered the "Fractional Indexing" system and "Optimistic Concurrency Control" logic.
- **Automated Verification:** Authored the complete testing framework, including unit, integration, and reliability test suites.